



Plant Disease Detection System

Steven An

Vikram Muruganandam

Youngjoon Park

Motivation

- Global food security is under constant threat.
- Every year, diseases destroy nearly 40% of global crop production.
- Traditional identification is slow and prone to human error



Fig 1. Cercospora Corn Leaf

Goal

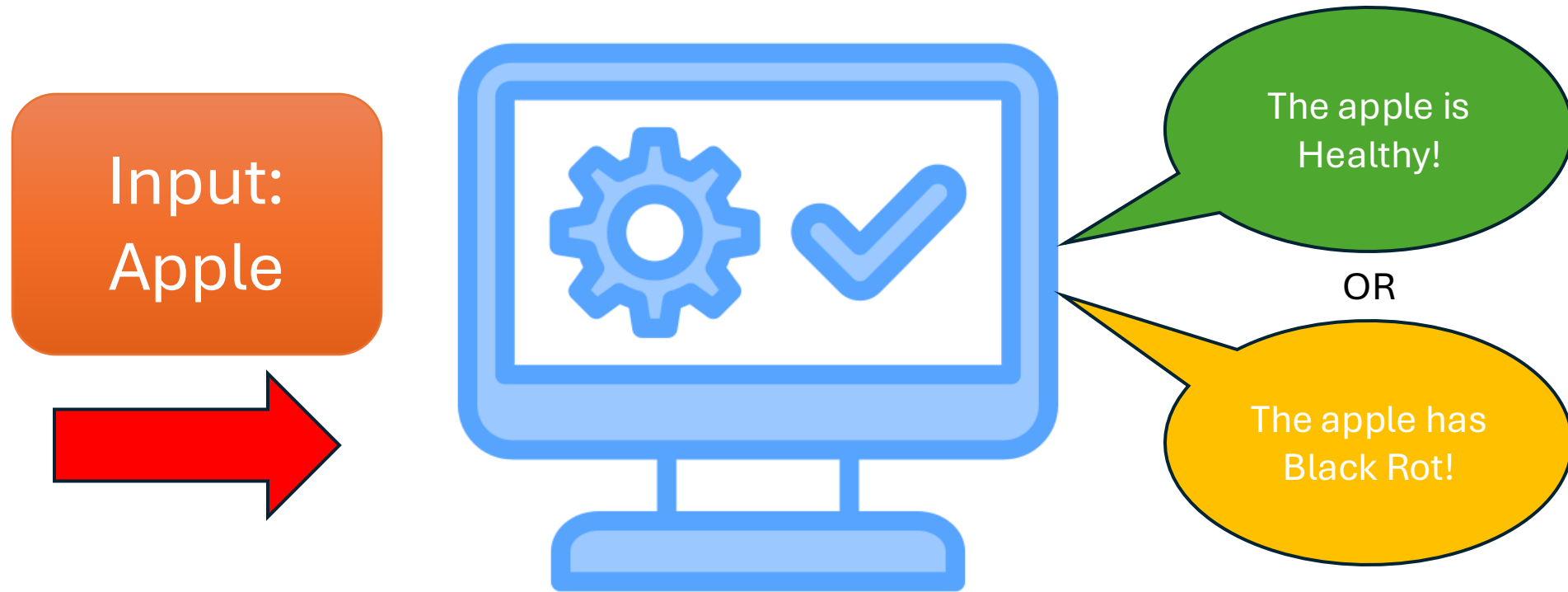


Fig 2. Goal Diagram

Can we build a multi class system to reliably classify whether the plant has disease and its type?

Method

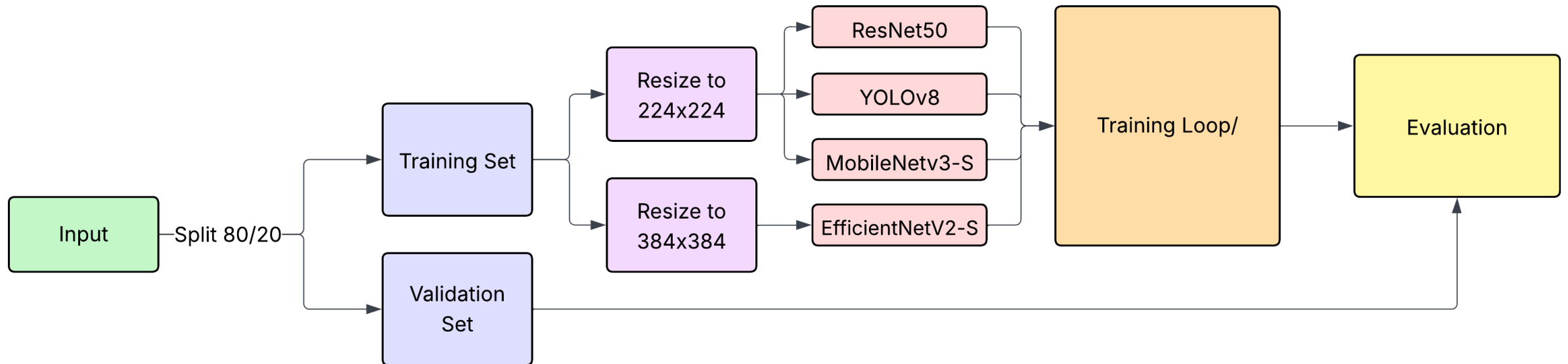


Fig. 3. 4-Stage System Diagram

Results

Models	Accuracy(%)
ResNet-50	98.52%
YOLOv8	99.8%
MobileNetV3-small	98.87%
EfficientNetV2-small	96%

Table 1. Accuracy Summary

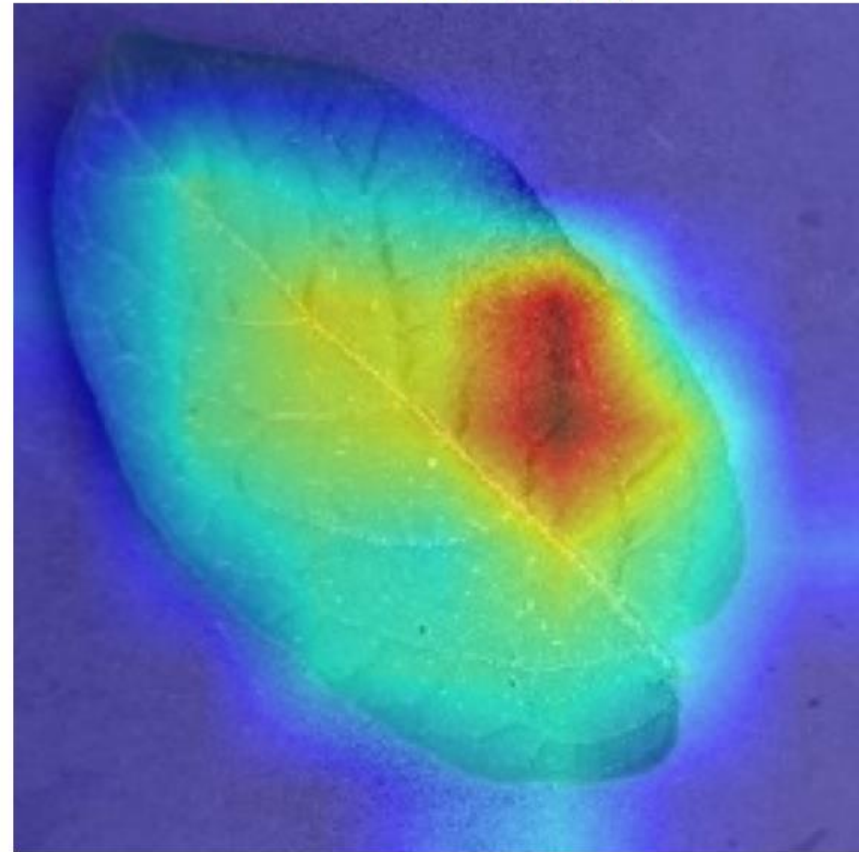
Grad-CAM Results

True Label: Potato__Late_blight

Original Image



Prediction: Potato__Late_blight



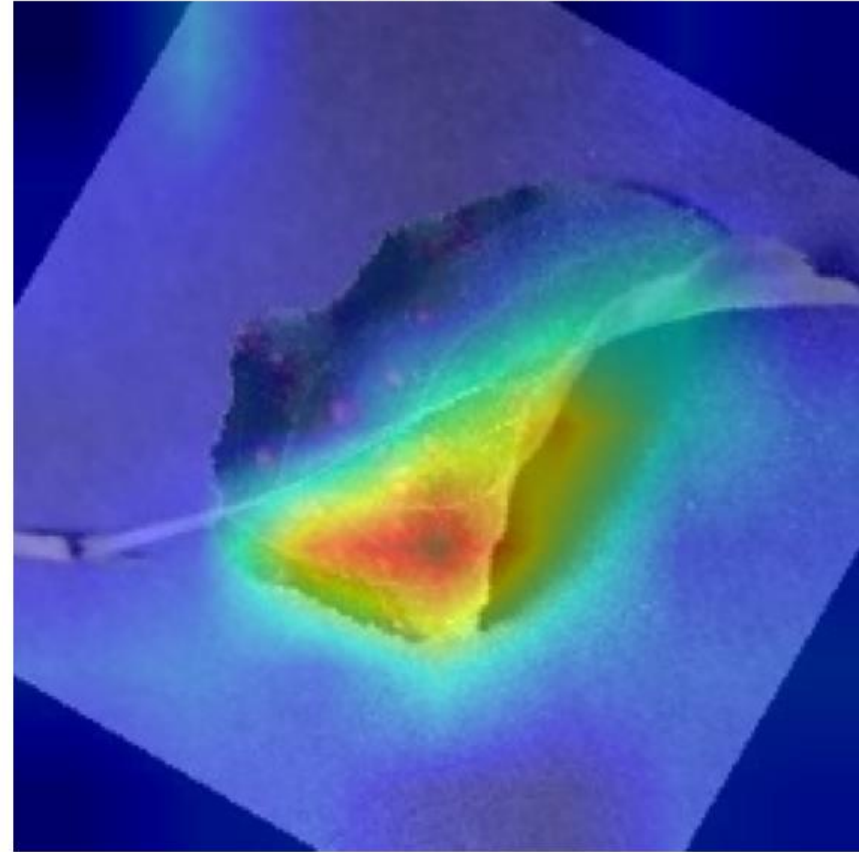
Grad-CAM Results

True Label: Apple__Cedar_apple_rust

Original Image



Prediction: Apple__Cedar_apple_rust



Grad-CAM Results

True Label: Apple__healthy

Original Image



Prediction: Apple__healthy

